

One State, Many Forward-Maximal Futures

Abstract

For the initial-value problem $\dot{x} = f(x)$, $x(0) = 0$ with f continuous, $f(0) = 0$ and $f > 0$ off the origin, the structure of the solution set is classical: it was characterised by Wallach [17] in 1948 and treated in general form by Binding [5] in 1979. This note takes one concrete member of that class, $f(x) = |x|^{2/3} + x^2$, and works it out completely.

The example has exactly one solution defined on all of $[0, \infty)$, namely $x \equiv 0$, and continuum-many solutions that admit no forward extension: for each waiting time $T \geq 0$ there is a solution that rests at the origin until T , departs, and reaches $+\infty$ at $T + T_*$, where the escape time $T_* < \infty$ is the same for every branch and is evaluated in closed form below. Two independent mechanisms produce the two counts. Failure of local Lipschitz uniqueness at the equilibrium permits a continuum of waiting-time departures, which is what makes the second count $\mathfrak{c}$; finite-time escape prevents every non-trivial branch from being global, which is what leaves the first count at 1. Neither mechanism alone separates them.

The distinction this illustrates — between uniqueness among globally defined solutions and uniqueness among solutions that merely cannot be extended forward — is not introduced here either. It is discussed for general relativity by Earman [8], Smeenk and Wüthrich [16] and Doboszewski [6], for Newtonian blow-up by Wilson [18], and within a broader formulation of indeterminism by Azhar and Namjoo [2]. What is offered here is an explicit worked example in which every quantity is available in closed form, together with one proposition (Proposition 10) locating where each of the two uniqueness verdicts is determined. **The example does not by itself establish that any physical theory is indeterministic.** What it shows is that a uniqueness verdict depends on which class of solutions is quantified over; Section 7 sets out the competing readings and adjudicates between none of them.

1 Introduction

Here is a question with a less obvious answer than it looks.

If an initial state admits exactly one solution that exists for all future times, does it follow that the state has a unique possible future history?

Not necessarily — not if histories are taken to be solutions that cannot be extended to any later time, rather than solutions defined at every future time. The two classes can differ, and where they do, the uniqueness verdict can differ with them. That distinction is the subject of this note; we claim no priority for it, nor for the mathematics on which it rests.

We use *history* throughout for a solution regarded as a candidate physical evolution. Nothing in Sections 2–5 depends on that identification; Section 7 is where it does work, and says so.

The mathematics is classical. For scalar autonomous equations the solution set through an equilibrium was characterised by Wallach [17] in 1948, and the general theory — the exact conditions for departure from an equilibrium, for escape to infinity in finite time, and for the structure of the solution set — is given by Binding [5]. The conceptual point is also familiar. In general relativity the maximal globally hyperbolic development of initial data is unique while extensions past a Cauchy horizon are not, and whether to restrict attention to the well-behaved class is a live question rather than a settled one [8, 16]. In Newtonian mechanics Wilson [18] argues that failures of global existence are descriptive gaps rather than failures of determinism, in explicit opposition to Earman and Norton. Azhar and Namjoo [2] argue that a uniqueness-based criterion misses the sense in which a terminating history is itself a failure of determination.

What follows is therefore expository. Its purpose is to put the distinction into a small explicit example and work it out in full: a single scalar equation, elementary integrals, a blow-up time in closed form, an explicit solution family. No general relativity, no N -body mechanics, no partial differential equations, no geometry.

2 The example

Throughout we study

$$\dot{x} = f(x), \quad x(0) = 0, \tag{1}$$

for a continuous $f: \mathbb{R} \rightarrow \mathbb{R}$.

Definition 1. A *solution* of (1) is a differentiable function $x: I \rightarrow \mathbb{R}$ on an interval $I \subseteq [0, \infty)$ with non-empty interior and $\min I = 0$, satisfying $\dot{x}(t) = f(x(t))$ throughout, with one-sided derivatives at any endpoint belonging to I .

We work forward from 0 until Appendix A. The standing hypotheses are

(H1) f is continuous, $f(0) = 0$, and $f(x) > 0$ for $x \neq 0$;

(H2) $\int_0^1 \frac{dx}{f(x)} < \infty$;

(H3) $\int_1^\infty \frac{dx}{f(x)} < \infty$.

The example is

$$f(x) = |x|^{2/3} + x^2.$$

Four properties matter, and all four are elementary.

Lemma 2. For $f(x) = |x|^{2/3} + x^2$:

(i) (H1) holds.

(ii) f is not locally Lipschitz at the origin, but is C^1 on $\mathbb{R} \setminus \{0\}$, so the failure sits at exactly one point.

(iii) The substitution $x = u^3$ gives $\frac{dx}{x^{2/3} + x^2} = \frac{3 du}{1 + u^4}$ for $x > 0$.

(iv) (H2) and (H3) both hold.

Proof. (i) $|x|^{2/3}$ is the composition of $x \mapsto |x|$ with $s \mapsto s^{2/3}$ on $[0, \infty)$, so f is continuous on $\mathbb{R}$; $f(0) = 0$; and $f(x) > 0$ for $x \neq 0$ since $|x|^{2/3} > 0$ and $x^2 \geq 0$.

(ii) $|f(x) - f(0)|/|x| = |x|^{-1/3} + |x| \rightarrow +\infty$ as $x \rightarrow 0$, so no Lipschitz constant works on any neighbourhood of the origin. Away from it, $f'(x) = \frac{2}{3} \operatorname{sgn}(x)|x|^{-1/3} + 2x$ is continuous.

(iii) For $x > 0$ put $x = u^3$, a C^1 increasing bijection of $(0, \infty)$. Then $dx = 3u^2 du$, $x^{2/3} = u^2$, $x^2 = u^6$, and

$$\frac{dx}{x^{2/3} + x^2} = \frac{3u^2 du}{u^2(1 + u^4)} = \frac{3 du}{1 + u^4}.$$

(iv) By (iii) the two integrals become $\int_0^1 3 du/(1 + u^4)$ and $\int_1^\infty 3 du/(1 + u^4)$. The first is finite because its integrand is continuous on the compact interval $[0, 1]$; the second because $3/(1 + u^4) \leq 3u^{-4}$ there, and u^{-4} is integrable on $[1, \infty)$. Both pieces are evaluated together in Proposition 3, where their sum is computed exactly. $\square$

The two halves of (iv) do different work: (H2) is what allows a solution to leave the equilibrium at all, and (H3) is what makes the departing solution reach $+\infty$ in finite time. Both facts are classical, and Section 6 says whose they are. Because the substitution rationalises the integrand completely, the total is available in closed form.

Proposition 3 (The blow-up time).

$$T_* := \int_0^\infty \frac{dx}{x^{2/3} + x^2} = 3 \int_0^\infty \frac{du}{1 + u^4} = 3 \cdot \frac{\pi/4}{\sin(\pi/4)} = \frac{3\sqrt{2}\pi}{4} \approx 3.3321622036.$$

Proof. The first equality is Lemma 2(iii); the second is the standard evaluation $\int_0^\infty du/(1 + u^n) = (\pi/n)/\sin(\pi/n)$ at $n = 4$; the third is arithmetic, since $\sin(\pi/4) = \sqrt{2}/2$. $\square$

The rationale for this f is exactly that and no more: the substitution $x = u^3$ rationalises the integrand completely, reducing the escape integral to the elementary $3 \int_0^\infty du/(1 + u^4)$ and giving T_* in closed form. We claim no uniqueness or optimality for the choice. Nothing in what follows depends on it; any member of the class would serve, and the results of Sections 3–4 are stated for the class rather than for this f .

3 The solution family

Define, for $x > 0$,

$$F(x) = \int_0^x \frac{du}{f(u)}. \quad (2)$$

Lemma 4. *Assume (H1)–(H3). Then F is a C^1 strictly increasing bijection from $(0, \infty)$ onto $(0, T_*)$ with $F(0^+) = 0$. Let $\varphi = F^{-1}$, extended by $\varphi(0) = 0$. Then φ is defined on $[0, T_*)$, is C^1 there including at 0, satisfies $\varphi' = f(\varphi)$ and $\varphi'(0^+) = 0$, and $\varphi(t) \rightarrow +\infty$ as $t \rightarrow T_*^-$.*

Proof. $F' = 1/f > 0$ is continuous on $(0, \infty)$, so F is C^1 and strictly increasing there; (H2) gives $F(0^+) = 0$ and (H3) gives $F(\infty^-) = T_* < \infty$, so F is onto $(0, T_*)$. Hence $\varphi = F^{-1}$ is defined on $(0, T_*)$, and $\varphi' = 1/F'(\varphi) = f(\varphi)$ there; $\varphi(t) \rightarrow +\infty$ as $t \rightarrow T_*^-$ because F is a bijection onto $(0, T_*)$.

It remains to treat $t = 0$, which is what lets the pieces glue in Proposition 5. Since $F(0^+) = 0$ and F is increasing, $\varphi(t) \rightarrow 0$ as $t \rightarrow 0^+$, so φ is continuous at 0 with $\varphi(0) = 0$. Then $\varphi'(t) = f(\varphi(t)) \rightarrow f(0) = 0$ by continuity of f . Now φ is continuous on $[0, t]$ and differentiable on $(0, t)$, so the mean value theorem gives $(\varphi(t) - \varphi(0))/t = \varphi'(\xi_t)$ for some $\xi_t \in (0, t)$; letting $t \rightarrow 0^+$ forces $\xi_t \rightarrow 0$ and hence $\varphi'(\xi_t) \rightarrow 0$. So the right derivative of φ at 0 exists and equals $0 = f(\varphi(0))$, and φ' is continuous at 0. $\square$

Proposition 5 (The waiting-time family). *For each $T \in [0, \infty)$ define*

$$x_T(t) = \begin{cases} 0, & 0 \leq t \leq T, \\ \varphi(t - T), & T < t < T + T_*, \end{cases}$$

and let $x_\infty \equiv 0$ on $[0, \infty)$. Then every x_T and x_∞ solves (1), and each x_T is C^1 on $[0, T + T_)$.*

Proof. On $[0, T)$ and on $(T, T + T_*)$ this holds by construction and Lemma 4. At $t = T$ the left derivative is 0, the right derivative is $\varphi'(0^+) = 0$ by Lemma 4, and $f(x_T(T)) = f(0) = 0$; so the two-sided derivative exists and equals $f(x_T(T))$. The join is therefore C^1 , not merely continuous. That x_∞ solves (1) is immediate from $f(0) = 0$. $\square$

In words: the solution rests at the equilibrium as long as it likes, leaves with zero slope at a time of its choosing, and then — after exactly T_* more, the same for every branch — runs off to $+\infty$.

One remark on why a one-parameter family is the whole of it. In the general structure theorem of Wallach and Binding, recorded in Section 6, a solution is assembled from the extremal one by adding two increasing parts supported on $f^{-1}(0)$: a continuous singular part and a jump part. Here $f^{-1}(0) = \{0\}$ is a single point, so the continuous singular part contributes nothing and the jump part reduces to a single number, the waiting time T . Theorem 7 proves completeness directly and does not rely on this; the remark is only to say where the one parameter comes from.

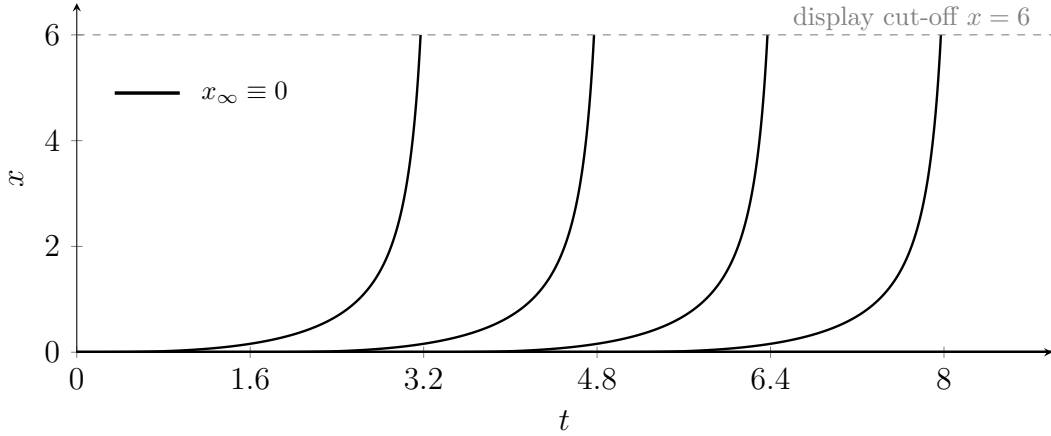


Figure 1: Four members of the family, departing at $T = 0, 1.6, 3.2, 4.8$. Each curve is φ , traced *for the drawing only* by sampling x and plotting $(T + F(x), x)$, so that F is evaluated and never inverted; the blow-up time is the analytic T_* of Proposition 3, and nothing here is offered as evidence for it. The curves are drawn only up to $x = 6$; that ceiling is graphical, and each branch in fact runs to $+\infty$ as t approaches $T + T_*$. Along the axis is x_∞ , the one solution still defined at every later time.

4 Globally defined versus forward-maximal

Two classes of forward solution, kept apart.

Definition 6. $\mathcal{H}_{\text{glob}}^+(0)$ is the set of solutions of (1) defined at *every* $t \geq 0$, that is, those with $I = [0, \infty)$.

$\mathcal{H}_{\text{max}}^+(0)$ is the set of solutions of (1) on intervals $I \subseteq [0, \infty)$ admitting no extension to a solution on a strictly larger $J \subseteq [0, \infty)$. We call these *forward-maximal*; Section 5 explains why the prefix is not optional.

Plainly $\mathcal{H}_{\text{glob}}^+(0) \subseteq \mathcal{H}_{\text{max}}^+(0)$: a solution on $[0, \infty)$ has nowhere further to go inside $[0, \infty)$. The main statement is that the two classes are nevertheless as far apart as they could be.

Theorem 7. Assume (H1)–(H3). Then, with $\mathcal{H}_{\text{max}}^+(0)$ the forward-maximal solutions of Definition 6 — not the two-sided maximal solutions of Section 5 —

$$\mathcal{H}_{\text{glob}}^+(0) = \{x_\infty\}, \quad \mathcal{H}_{\text{max}}^+(0) = \{x_T : T \in [0, \infty)\} \cup \{x_\infty\},$$

so that $|\mathcal{H}_{\text{glob}}^+(0)| = 1$ while $|\mathcal{H}_{\text{max}}^+(0)| = \mathfrak{c}$.

Proof. We argue in eight steps.

Step 1 (monotonicity). Since $f \geq 0$ everywhere, every solution has $\dot{x} \geq 0$ and so is non-decreasing; as $x(0) = 0$, every solution is non-negative.

Step 2 (the zero set). Let x be a solution on I and put $Z = \{t \in I : x(t) = 0\}$. Then Z is closed in I by continuity, and is an interval: if $t_1 < t < t_2$ with $t_1, t_2 \in Z$ then $0 = x(t_1) \leq x(t) \leq x(t_2) = 0$ by Step 1. Also $0 \in Z$. Set $T = \sup Z \in [0, \infty]$.

Step 3 (each x_T is forward-maximal). By Lemma 4, $x_T(t) \rightarrow +\infty$ as $t \rightarrow (T + T_*)^-$. Any $J \subseteq [0, \infty)$ strictly containing $I = [0, T + T_*)$ and admitting a solution extending x_T would have to contain the point $T + T_*$, at which the extension would have to be finite and continuous — impossible. So $x_T \in \mathcal{H}_{\max}^+(0)$. Likewise $x_\infty \in \mathcal{H}_{\text{glob}}^+(0) \subseteq \mathcal{H}_{\max}^+(0)$.

Step 4 (no forward-maximal solution has a compact domain). Suppose $x \in \mathcal{H}_{\max}^+(0)$ has $I = [0, b]$ with $b < \infty$. Then $x(b)$ is finite and f is continuous, so Peano's existence theorem supplies a solution y of $\dot{y} = f(y)$ on some $[b, b + \delta]$ with $y(b) = x(b)$. Concatenating x and y gives a solution on $[0, b + \delta]$ properly extending x , contradicting maximality. Hence every I arising in $\mathcal{H}_{\max}^+(0)$ is right-open or is $[0, \infty)$.

Step 5 (the case $T = \infty$). If $T = \infty$ then Z is unbounded, and being an interval containing 0 it equals I ; so $x \equiv 0$ on I . If $I \neq [0, \infty)$ then extending by 0 gives a solution on a strictly larger subinterval of $[0, \infty)$, contradicting maximality. So $I = [0, \infty)$ and $x = x_\infty$.

Step 6 (the case $T < \infty$). First, $T \in I$. Otherwise I is an interval containing $[0, T)$ but not T , hence $I \subseteq [0, T)$ and $x \equiv 0$ on I , so $x(t) \rightarrow 0$ as $t \uparrow T$. Setting $x(T) = 0$ extends x continuously to $[0, T]$, and the extension solves (1) at T because $f(0) = 0$; it is then extended further by the zero solution. This contradicts maximality. (Step 4 handles the general version of the same situation: at a compact right endpoint at which the solution is finite, Peano's existence theorem gives a local solution from that value, which concatenates to a proper extension. Here the value is the equilibrium, so the extension is the zero solution and no appeal to Peano is needed.)

For $t \in I$ with $t > T$ we have $x(t) \neq 0$ by the definition of T as $\sup Z$, and $x(t) \geq x(T) = 0$ by Step 1; hence $x(t) > 0$. On $I \cap (T, \infty)$ therefore $\dot{x} = f(x) > 0$, so x is strictly increasing there and $\dot{x}/f(x) = 1$. Integrating from s to t with $T < s < t$ and substituting $u = x(\sigma)$ — legitimate since x is a C^1 strictly increasing bijection from $[s, t]$ onto $[x(s), x(t)]$ — gives

$$t - s = \int_s^t \frac{\dot{x}(\sigma)}{f(x(\sigma))} d\sigma = \int_{x(s)}^{x(t)} \frac{du}{f(u)} = F(x(t)) - F(x(s)).$$

Letting $s \downarrow T$ and using $x(s) \rightarrow x(T) = 0$ together with $F(0^+) = 0$ yields $F(x(t)) = t - T$, that is $x(t) = \varphi(t - T)$. Since F takes values in $(0, T_*)$ this forces $t - T < T_*$, so $I \subseteq [0, T + T_*)$. By Step 4 the domain is not compact, and by maximality together with Step 3 it cannot be a proper subinterval of $[0, T + T_*)$; hence $I = [0, T + T_*)$ and $x = x_T$.

Step 7 (the cardinality of $\mathcal{H}_{\max}^+(0)$). Steps 3, 5 and 6 together give the stated description of $\mathcal{H}_{\max}^+(0)$. For the lower bound, $T \mapsto x_T$ is injective: if $T_1 < T_2$ then on the interval $(T_1, T_2]$, of positive length, $x_{T_1} = \varphi(\cdot - T_1) > 0$ while $x_{T_2} = 0$, so the two functions differ. Moreover $x_\infty \neq x_T$ for every $T < \infty$, since $x_\infty \equiv 0$ while $x_T > 0$ on $(T, T + T_*)$; so the listed solutions are pairwise distinct and $|\mathcal{H}_{\max}^+(0)| \geq |[0, \infty)| = \mathfrak{c}$.

For the upper bound, every solution is a continuous real-valued function on a subinterval of $[0, \infty)$. There are $\mathfrak{c}$ such subintervals, and a continuous function on an interval is determined by its values on a countable dense subset, so each interval carries at most $\mathfrak{c}^{\aleph_0} = \mathfrak{c}$ of them. Hence $|\mathcal{H}_{\max}^+(0)| \leq \mathfrak{c} \cdot \mathfrak{c} = \mathfrak{c}$. Combining the bounds, $|\mathcal{H}_{\max}^+(0)| = \mathfrak{c}$.

Step 8 (the globally defined solutions). Let $x \in \mathcal{H}_{\text{glob}}^+(0)$, so $I = [0, \infty)$. A solution on $[0, \infty)$ admits no extension inside $[0, \infty)$, so $\mathcal{H}_{\text{glob}}^+(0) \subseteq \mathcal{H}_{\max}^+(0)$ and x is on the list above. It is not any x_T with $T < \infty$: by Lemma 4 such a branch satisfies $x_T(t) \rightarrow +\infty$ as $t \uparrow T + T_*$, so it is defined only on the bounded interval $[0, T + T_*)$ and in particular not at $t = T + T_*$. Every non-zero branch therefore reaches $+\infty$ after the finite additional time T_* and fails to be global. The only remaining possibility is $x = x_\infty$, which is indeed defined on all of $[0, \infty)$. Hence $\mathcal{H}_{\text{glob}}^+(0) = \{x_\infty\}$ and $|\mathcal{H}_{\text{glob}}^+(0)| = 1$. $\square$

Remark 8 (The counts do not depend on the solution concept). Definition 1 asks for a differentiable solution; Binding [5] works instead with absolutely continuous x satisfying $\dot{x} = f(x)$ almost everywhere. The two give the same sets here. If x is absolutely continuous with $\dot{x} = f(x)$ a.e., then $x(t) = x(0) + \int_0^t f(x(s)) ds$, and $f \circ x$ is continuous, so x is C^1 and solves (1) in the classical sense. So Theorem 7 is not an artefact of the regularity demanded of solutions.

Remark 9 (Each hypothesis moves one count). Without (H2) no branch could leave the origin, and both counts would be 1; without (H3) every x_T would be defined on all of $[0, \infty)$, and both counts would be $\mathfrak{c}$. Only the conjunction separates them. The margin is also the largest available, and both ends of it are fixed. At the upper end, by Step 7 no solution set of (1) can exceed $\mathfrak{c}$. At the lower end, (H1) gives $f(0) = 0$, so $x_\infty \equiv 0$ on $[0, \infty)$ satisfies Definition 1 and lies in $\mathcal{H}_{\text{glob}}^+(0)$; hence $|\mathcal{H}_{\text{glob}}^+(0)| \geq 1$ whenever (H1) holds. The pair $(1, \mathfrak{c})$ of Theorem 7 therefore realises the smallest global count together with the largest forward-maximal one, and no instance of (1) satisfying (H1) can separate the two counts further.

The remark can be stated exactly, and stating it exactly locates *where* each of the two verdicts is determined. The next proposition does no more than read that off the three cases; its ingredients are the classical dichotomies attributed in Section 6, and we claim no priority for it.

Proposition 10 (Where each verdict is determined). *Assume (H1), but not necessarily (H2) or (H3). Then*

$$|\mathcal{H}_{\max}^+(0)| = 1 \iff \neg(\text{H2}), \quad |\mathcal{H}_{\text{glob}}^+(0)| = 1 \iff \neg(\text{H2}) \vee (\text{H3}).$$

The condition on the left is determined by the restriction of f to any neighbourhood of the origin. The condition on the right is not determined by the restriction of f to any bounded set.

Proof. Three cases exhaust the possibilities.

Case $\neg(\text{H2})$. We claim $x \equiv 0$ is the only solution. This is the departure criterion recorded in Section 6; the direct argument is short enough to give, and keeps the

proposition self-contained. Suppose some solution x on I had $x(t_0) > 0$ for some $t_0 \in I$. Steps 1 and 2 of Theorem 7 use only (H1), so x is non-decreasing and non-negative and $Z = \{t \in I : x(t) = 0\}$ is an interval containing 0. Since $t_0 \notin Z$, and Z is an interval containing 0, we get $Z \subseteq [0, t_0]$; hence $T := \sup Z \leq t_0 < \infty$ and $[0, T] \subseteq Z$. As $[0, t_0] \subseteq I$ we have $T \in I$, and $x(T) = 0$ by continuity; moreover $T < t_0$, since $x(T) = 0 < x(t_0)$. And $x(s) > 0$ for every $s \in (T, t_0]$, since such an s lies outside Z while $x \geq 0$. The computation in Step 6 of Theorem 7 — which uses only (H1) and the strict positivity of x on $I \cap (T, \infty)$, not (H2) or (H3) — gives

$$t_0 - s = \int_{x(s)}^{x(t_0)} \frac{du}{f(u)} \quad \text{for every } s \in (T, t_0).$$

Let $s \downarrow T$. The left-hand side tends to $t_0 - T < \infty$. On the right, $x(s) \rightarrow x(T) = 0$ by continuity and $1/f > 0$ on $(0, \infty)$, so the integral increases to $\int_0^{x(t_0)} du/f(u)$. That limit is $+\infty$. Indeed, put $c = \min(1, x(t_0)) > 0$; all three terms in $\int_0^1 du/f = \int_0^c du/f + \int_c^1 du/f$ are non-negative, the left side is $+\infty$ by $\neg(\text{H2})$, and $\int_c^1 du/f$ is finite because $1/f$ is continuous on the compact interval $[c, 1] \subset (0, \infty)$; so $\int_0^c du/f = +\infty$, and $\int_0^{x(t_0)} du/f$ dominates it. This contradicts $t_0 - T < \infty$. Hence no solution takes a positive value, and with Step 1 every solution is identically 0 on its domain. The extension argument inside Step 5 — which also uses only (H1), and does not depend on that step's case hypothesis — then forces $I = [0, \infty)$ for a forward-maximal solution: a solution identically 0 on a proper subinterval of $[0, \infty)$ is properly extended by the zero solution. So $\mathcal{H}_{\max}^+(0) = \mathcal{H}_{\text{glob}}^+(0) = \{x_\infty\}$ and both counts are 1.

Case (H2) and (H3). Theorem 7 gives $|\mathcal{H}_{\max}^+(0)| = \mathfrak{c}$ and $|\mathcal{H}_{\text{glob}}^+(0)| = 1$.

Case (H2) and $\neg(\text{H3})$. Then F is a bijection from $(0, \infty)$ onto $(0, \infty)$, so φ is defined on all of $[0, \infty)$ and every x_T is global; the argument of Theorem 7 is otherwise unchanged, and $\mathcal{H}_{\max}^+(0) = \mathcal{H}_{\text{glob}}^+(0) = \{x_T : T \in [0, \infty)\} \cup \{x_\infty\}$, of cardinality $\mathfrak{c}$.

Reading the two counts off the three cases gives the two equivalences.

The locality claims. First, (H2) is determined by the restriction of f to $(0, \delta)$ for every $\delta \in (0, 1]$: by (H1) $f > 0$ on the compact interval $[\delta, 1] \subset (0, \infty)$, so $1/f$ is continuous and hence integrable there, and therefore $\int_0^1 du/f$ and $\int_0^\delta du/f$ converge together. For $\delta > 1$ the claim is immediate, since then $(0, \delta) \supseteq (0, 1)$. Equivalently: if two functions satisfying (H1) agree on some neighbourhood of the origin, then either both satisfy (H2) or neither does.

Second, (H3) is not determined by the restriction of f to any bounded set. Every bounded set is contained in $[-M, M]$ for some $M > 0$, so it suffices to produce, for each $M > 0$, two functions satisfying (H1) that agree on the whole of $[-M, M]$ and disagree about (H3). One pair agreeing on $[-1, 1]$ would settle only the case $M = 1$. Take $f(x) = |x|^{2/3} + x^2$ as in Section 2, fix $M > 0$, and put

$$f_M^\flat(x) = \begin{cases} f(x), & |x| \leq M, \\ f(M) |x|/M, & |x| \geq M. \end{cases}$$

Continuity. Each branch is continuous where it is defined, and at $|x| = M$ the second reads $f(M)M/M = f(M)$, so the branches agree there; hence f_M^b is continuous on $\mathbb{R}$.

(H1). The hypothesis asks for continuity, the value 0 at the origin, and positivity off the origin, and for nothing else. Continuity is just shown; $f_M^b(0) = f(0) = 0$, since $0 \leq M$; for $0 < |x| \leq M$ we have $f_M^b(x) = f(x) > 0$; and for $|x| \geq M$ we have $f_M^b(x) = f(M)|x|/M > 0$, because $f(M) = M^{2/3} + M^2 > 0$ and $M > 0$.

(H2) *is preserved.* f_M^b agrees with f on $[-\min(M, 1), \min(M, 1)]$, a neighbourhood of the origin, so by the paragraph above it satisfies (H2) exactly when f does — which f does, by Lemma 2(iv).

(H3) *fails.* Put $N = \max(M, 1)$. On $[N, \infty)$ we have $f_M^b(x) = f(M)x/M$, so $\int_N^\infty dx/f_M^b = (M/f(M)) \int_N^\infty dx/x = \infty$, while $\int_1^N dx/f_M^b$ is finite, its integrand being continuous on the compact interval $[1, N] \subset (0, \infty)$. Hence $\int_1^\infty dx/f_M^b = \infty$.

Agreement. By construction $f_M^b = f$ on the whole of $[-M, M]$.

So f satisfies (H2) and (H3), while f_M^b satisfies (H2) and not (H3), and the two agree on $[-M, M]$. By the equivalences just proved, $|\mathcal{H}_{\text{glob}}^+(0)| = 1$ for f and $|\mathcal{H}_{\text{glob}}^+(0)| = \mathfrak{c}$ for f_M^b , whereas $|\mathcal{H}_{\text{max}}^+(0)| = \mathfrak{c}$ for both. Since $M > 0$ was arbitrary and every bounded set lies in some $[-M, M]$, no bounded set determines (H3). $\square$

So the two counts are not determined in the same place. Whether one solution or continuum-many cannot be extended forward is settled by the law near the state; whether one solution or continuum-many are defined at every later time is not, and no amount of information about the law near the state settles it.

One remark, to forestall a misreading. Proposition 10 locates each condition in the coordinate x , in which the initial state and both solution classes are posed; it does not say that finite-time escape has no local description at all. The reciprocal substitution recorded in Section 6 carries $\dot{x} = f(x)$ into $\dot{\alpha} = -\hat{f}(\alpha)$ with $\hat{f}(\alpha) = \alpha^2 f(1/\alpha)$, and Binding's Lemma 6.3 records that escape to infinity for f corresponds to non-uniqueness at the origin for $\hat{f}$. Read in α instead of in x , (H3) is again a condition at the origin; the contrast drawn by the proposition is between two conditions read in two different coordinates, not between one that has a local reading and one that has none.

5 Three senses of “maximal”

The word carries at least three meanings in the surrounding literature, and conflating them produces false statements about this example. We separate them once and then keep the prefix.

- **Forward-maximal:** no extension within $[0, \infty)$. These are the x_T , on $[0, T + T_*)$ — the class counted in Theorem 7.
- **Maximal (standard):** no extension to any larger interval of $\mathbb{R}$. *No x_T is maximal in this sense.* Setting $x(t) = 0$ for $t < 0$ always extends it, since $f(0) = 0$. Appendix A classifies the genuinely maximal solutions.

- **Maximal (extremal):** the *largest* solution pointwise on the common interval of existence — the qualification matters here, since the candidates have different domains. That solution is x_0 , which departs immediately; the smallest in the same sense is x_∞ . Hartman [11, ch. III §2] uses the word in this sense.

So $[0, T + T_*)$ is a forward-maximal interval and not a maximal interval, and $\mathcal{H}_{\max}^+(0)$ is not a set of maximal solutions.

6 What is classical, and where

The results above rest on classical theory, and we claim no priority for any of them. The attributions below are specific rather than general, because a general gesture at “the classical theory” would leave a reader unable to check them.

Wallach [17], 1948. Characterised *all* solutions of $\dot{x} = f \circ x$, $x(0) = 0$ for continuous f . Binding reports this at [5, p. 184] — “Wallach characterized all solutions of (1.1)” — and credits the monotonicity lemma and the structure theorem to two sections of it, which he cites as Wallach’s §2 and §7 respectively. *We know this paper only through Binding and have not read it; the two section numbers are Binding’s citations, reproduced here, and we have not checked them against Wallach.*

Binding [5, §3, pp. 186–188]. The machinery of Theorem 7. Lemma 3.2: every solution is monotonic, which Binding credits to Wallach [17] for continuous f — our Step 1. Theorem 3.3: the separation identity $\lambda([s, t] \setminus \dot{x}^{-1}(0)) = \int_{x(s)}^{x(t)} 1/f$ in measure-theoretic form, of which the computation in Step 6 is the elementary case. Theorem 3.5: necessary conditions for local existence, reducing for continuous f with $x > 0$, in his words, to “ $f > 0$ and $1/f$ integrable, as given by Wallach”.

Binding [5, §§5–6, pp. 190–193]. Theorem 5.3 is the uniqueness criterion, in the Osgood [15] and Kamke line — Binding names both on the same page — and see Hartman [11, ch. III §6], where Theorem 6.1 is Kamke’s and Corollary 6.2 is Osgood’s. Fjordholm [9], whose own subject is Filippov uniqueness for discontinuous fields, restates the continuous case as “it was shown by Binding that the solution is unique if and only if b satisfies the so-called Osgood condition at all zeroes of b ”. Binding’s case (6.5) corresponds to (H2) and (H3) together, not to (H3) alone: the integral in that case runs from the equilibrium, so its convergence near the origin is (H2) and its convergence at infinity is (H3). With (H2) standing, as it does throughout this note, it is (H3) that does the work. Of that case Binding writes, “we have the phenomenon of escape to infinity for at least one solution”; Binding’s case (6.2), where the integral diverges, is the complementary one in which the branch survives on the whole half-line. Lemma 6.3 is a duality: the reciprocal substitution turns escape to infinity into non-uniqueness at the origin, since $x = 1/\alpha$ carries $\dot{x} = f(x)$ into $\dot{\alpha} = -\hat{f}(\alpha)$ with

$$\hat{f}(\alpha) = \alpha^2 f(1/\alpha), \tag{3}$$

and, up to the orientation of the two integrals, $dx/f(x) = d\alpha/\hat{f}(\alpha)$, so $\int^\infty 1/f$ and $\int_0 1/\hat{f}$ converge together. The form (3) is Binding’s own: it is the definition printed immediately

before his Lemma 6.3, and his inversion identity — the display numbered (6.6) in the JDE printing — is the computation just given. The observation that (H2) and (H3) are the two ends of one integral is likewise his, and he credits awareness of it to Wintner, to Brauer and Sternberg, and to Yoshizawa.

Binding [5, Lemma 7.5, p. 194]. The structure theorem: every solution is obtained from the extremal one by inserting constant segments at zeros of f . With $f^{-1}(0) = \{0\}$ that is precisely the waiting-time family of Proposition 5. Binding notes that Wallach’s proof for continuous f carries over.

Hartman [11, ch. III §5 Thm. 5.1, p. 30]. A theorem of Wintner: if $\int^\infty du/\psi = \infty$ and $|f| \leq \psi$, solutions exist on the whole strip; finite-time escape requires the negation. Exercise 6.4 (p. 33), attributed in the chapter Notes to Wallach, states the departure criterion as an exact dichotomy; it is also the standard source for the textbook waiting-time examples $\dot{x} = x^{1/3}$ and $\dot{x} = \sqrt{|x|}$, whose continuum of wait-then-depart solutions is the classical Peano phenomenon and is not due to any recent author. The general fact that non-uniqueness forces continuum-many solutions — our Step 7, in far greater generality — is Kneser’s theorem, [11, ch. II §4 Thm. 4.1].

Drivas and Mailybaev [7] (preprint 2018; published 2021). Cited here for continuation past a non-Lipschitz singularity and the regularisation that selects among the continuations — the context their paper is actually about — and not as the origin of the waiting-time family, which is classical. In passing they record it for $\dot{x} = x^{1/3}$: “a continuum infinity of solutions, which stay at the origin until an arbitrary time t_1 and peel [sic] off.” There the branches are global, so no gap between the two counts arises.

Berndl et al. [4], 1995. Use “maximal (non-extendible) solution” against “global existence” as basic vocabulary in a foundational setting, and list escape to infinity in finite time among the ways a solution may cease to exist.

Given Lemma 7.5 and case (6.5) together, Theorem 7 is close to immediate, with one gap that has to be closed by hand: Binding’s §7 carries the standing hypothesis that all solutions are positive for $t > 0$, which excludes x_∞ — the one solution whose status in $\mathcal{H}_{\text{glob}}^+(0)$ is the point at issue — so the zero solution and the bookkeeping of Steps 3 and 4 are not supplied by his Lemma 7.5 as printed. The self-contained proof above is written out because it is short and because of that gap, not because the result is new. The wider uniqueness and non-uniqueness literature is surveyed in Agarwal and Lakshmikantham [1].

On the particular choice of f : we have not found a prior publication using $|x|^{2/3} + x^2$ to illustrate this point, though the search behind that remark is bounded and we make no priority claim. What can be said without qualification is that the substitution puts the blow-up time in closed form; we claim no advantage for the choice beyond that.

7 Bearing on determinism

The example admits at least three readings. **None of them is introduced here.** We set them out with attributions and do not adjudicate between them.

First, what is being tested. We do not propose a definition of determinism; we borrow one already in use, so that the readings below have something definite to disagree about. Doboszewski [6, p. 2] adopts the criterion of Butterfield (1989) — his DM2 — under which a theory is deterministic for a given kind of region when any two models agreeing on a region of that kind agree throughout, so that, in his gloss, “to find a witness of indeterminism one needs merely to find two models that are isomorphic in region S but are not isomorphic overall — just as standard Laplacian intuitions demand.” Transposed to the present setting with all due looseness, the region is the initial state $x(0) = 0$, the models are the solutions through it, and the question is whether agreement there forces agreement throughout. **Everything then turns on what “throughout” ranges over** — which is precisely the choice Theorem 7 makes visible. Other criteria are in circulation and give different verdicts on this example; we take no stand on which is correct, and the example is not evidence for any of them.

Müller and Placek [13] separate three families of definition — one reading determinism off a theory’s defining differential equations, one off mappings between temporal realizations, one off branching models — and the two readings below sit inside that survey rather than outside it. Their equation-based family draws the present distinction directly, observing that “it is global uniqueness that naturally corresponds to determinism” while the classical theorems supply only local solutions, so that “[n]on-extendability then points to a possible failure of determinism”; and their branching definition counts *maximal chains* rather than total ones, which gives it Reading 2’s shape rather than Reading 1’s. They apply all three criteria to Norton’s dome [14], where the wait-then-depart branches are global and the gap counted in Theorem 7 therefore does not arise; we do not suggest that they consider the present equation, and nothing below is offered as a reading of their view. One further observation of theirs bears on the count itself. If the state has been at the equilibrium for all earlier times, they note, the branches are related by a time-translation symmetry, “which suggests that all these solutions represent a single possibility”; a footnote adds that this collapse fails when there is a first moment at which the state is set, since the branches are then “not time-translation symmetric”. Here Definition 1 imposes $\min I = 0$, so it is the second case that obtains, and the branches of Proposition 5 are not time-translates of one another within $[0, \infty)$.

Reading 1: the global-solution criterion

If a physical history must be defined at every time, the branches x_T are not histories, the state $x = 0$ has exactly one future, and the example says something about an initial-value problem and nothing about determinism. This is close to the position Wilson [18, p. 182] defends for Newtonian mechanics: failures of global existence are “descriptive holes”, and “it is as descriptive holes such issues should be addressed, *not as failures of determinism*.” He names Earman and Norton as holding the contrary view and calls their classification “terminologically unfortunate”.

Reading 2: the forward-maximal-history criterion

If a solution that cannot be extended forward counts as a history even though it ends, the same state has continuum-many futures, and requiring totality looks like a stipulation that secures determinism rather than a discovery that it holds. The structure of this objection is familiar from general relativity, where the maximal globally hyperbolic development of initial data is unique but, in Earman’s phrase [8, §6.4], often “not maximal simpliciter”, and the larger extensions are not unique. Smeenk and Wüthrich [16] organise their treatment around whether global hyperbolicity should be imposed by fiat, established from weaker assumptions, or justified from outside the theory — the preprint’s §5 is titled “Global Hyperbolicity by Fiat” — and note that their analysis “runs parallel to recent discussions of determinism in Newtonian particle mechanics.” The parallel is one of form and not of kind: in general relativity the restriction that secures uniqueness is a restriction on which models count as models of the theory, whereas the restriction at issue here is on the temporal domain a solution must have.

A closer analogue, though only an analogue, is Doboszewski’s discussion [6, p. 12] of spacetimes that simply stop — Minkowski space with a spacelike slice and its future removed — for which he coins *Doomsday indeterminism*, remarking that “there is no physical reason for spacetime to come to an end abruptly in this way.” The comparison should be kept limited and it is not an equivalence: his truncations are produced by excising the future by hand, whereas here the finite endpoint is generated by the equation. That difference bears on his own desideratum, paraphrased from Earman and Norton, that “determinism may fail, but (one hopes) if it fails, it should fail because of reasons of physics, not because of a clever mathematical trick” [6, p. 2]. We claim no priority over that discussion and do not suggest the two constructions are of a kind.

Reading 3: termination as its own failure

A third position holds that the uniqueness-based formulation is the wrong test in the first place. Azhar and Namjoo [2] propose that indeterminism arises when future states are “(i) not unique . . . ; (ii) not specified; or (iii) ‘incoherent’”, and argue that a Laplacian criterion based on non-uniqueness alone “does not comprehensively capture the relevant sense of a lack of determination”. On such a view the branches here witness a failure of determination by ending, independently of how many of them there are.

What happens at $T + T_*$

The mathematical fact is bare: $x_T(t) \rightarrow +\infty$ as $t \uparrow T + T_*$, and no continuation past that time exists. What that *is* remains open, and at least three answers are available: (a) the branch is a history that terminates at a finite time; (b) the divergence signals that the model has broken down and is no longer describing anything; (c) $+\infty$ is not a state, so the solution leaves the state space and simply ceases to exist.

The mathematics is indifferent: Theorem 7 holds under all three, because it is a statement about a set of solutions and not about what solutions represent. The

determinism verdict is not indifferent. The example bears on determinism only under (a), or under a criterion on which failure of continuation counts against determinism however the failure is described. **We therefore make no unconditional claim that any physical theory is indeterministic.** Under (b) or (c) the example remains exactly what Theorem 7 says it is, and nothing more.

As to the positions, the example adds nothing. What it adds is a test case in which the three readings give visibly different verdicts about one state of one system, with no idealisation to argue over: the equation is scalar and autonomous, f is continuous everywhere and C^1 away from one point, and every quantity is exact. One point is common to all three readings: the distinction is not dissolved by demanding more regularity. The Lipschitz failure sits at the equilibrium alone, and the branches terminate because the equation is superlinear at infinity, which is ordinary behaviour.

Proposition 10 bears on the three readings, and it is worth being exact about how. It is a statement about two classes of solutions of one equation, not a verdict about the world, and we claim no priority for it. What it supplies is this: a uniqueness-based criterion that quantifies over the solution classes studied in this paper should make explicit whether its uniqueness verdict at a state is fixed by the law in a neighbourhood of that state. For $\mathcal{H}_{\max}^+(0)$, the class Reading 2 quantifies over, the answer is yes: $|\mathcal{H}_{\max}^+(0)| = 1$ exactly when (H2) fails, and (H2) is determined by f near the origin. For $\mathcal{H}_{\text{glob}}^+(0)$, the class Reading 1 quantifies over, the answer is no: $|\mathcal{H}_{\text{glob}}^+(0)| = 1$ exactly when (H2) fails or (H3) holds, and (H3) is a condition on the tail of f . The witnesses in the proof make the difference concrete: for each $M > 0$, f_M^b agrees with f on all of $[-M, M]$ and differs from it only in the tail, and that change alone leaves the forward-maximal count at $\mathfrak{c}$ while moving the global count from 1 to $\mathfrak{c}$.

The three readings pay for this differently, and the proposition ranks nothing. On Reading 1 the verdict at a state depends on the law arbitrarily far from that state; this is a cost to be stated and answered, not a refutation of the reading. Reading 2 does not pay that cost and pays another: on it a forward-maximal branch that ends counts as a distinct future, which is what Reading 1 declines to allow. Reading 3 declines the uniqueness-based formulation altogether, so the proposition does not reach it; the cost there is that whatever settles the verdict must be supplied from outside that formulation. We adjudicate between none of the three readings.

8 What this note contributes

The phenomenon is classical and the distinction has substantial precedent, so the note claims no priority for the theory it draws on and offers no new criterion of determinism. Two things are offered. The first is the worked example itself: a scalar autonomous equation in which both failure modes occur together, every quantity is exact, and each step can be checked without general relativity, N -body mechanics or partial differential equations. Where the two failure modes are usually illustrated separately — Norton’s dome [14] for non-uniqueness; the non-collision singularities of Xia [19] for finite-time escape — here they occur in one equation. The second is Proposition 10, which says

where each of the two verdicts is determined: under (H1), the count of forward-maximal histories is fixed by f in any neighbourhood of the initial state, while the global count is not fixed by f on any bounded set. The ingredients of that proposition are the classical dichotomies of Section 6, and we claim no priority for it.

Limitations. Everything above concerns a scalar autonomous equation; the monotonicity argument of Step 1 uses the scalar order structure and does not survive to systems without new work. Nothing here is a theorem about Newtonian N -body dynamics. Appendix B gives a one-dimensional mechanical system exhibiting the mechanism, but it realises the mechanism, not this f . The interpretive question of Section 7 is left open, as it is in the literature cited there.

Conclusion. The example shows, in a simple setting, why a statement about uniqueness must specify the class of solutions being quantified over. Uniqueness among globally defined solutions and uniqueness among solutions that cannot be extended forward are different properties of the same law, and here they disagree at the same state, by the largest margin available.

A Two-sided time

The forward account says nothing about $x < 0$, and for this f , which is even, the negative half-line is populated: solutions can also arrive at the origin from below in finite time. Accounting for them exhibits solutions maximal in the standard sense and confirms that they are not the objects counted in Theorem 7.

Write $S_* = \int_{-\infty}^0 du/f(u)$. Two conditions are involved, and they can fail independently in at least one direction; the converse witness is not given here:

(H4a) $\int_{-1}^0 du/f < \infty$ — finite-time access to the equilibrium from below;

(H4b) $\int_{-\infty}^{-1} du/f < \infty$ — finite-time escape to $-\infty$ in backward time.

For an even f both follow from (H2) and (H3). In general they need not: Example 11 exhibits (H4b) without (H4a), so (H4a) does not follow from the conditions imposed on the positive half-line. We do not give the witness in the other direction.

Example 11 ((H4b) without (H4a)). Take $f(x) = x^{2/3} + x^2$ for $x \geq 0$ and $f(x) = |x| + x^2$ for $x < 0$. This is continuous, and the forward conditions hold because the positive half-line is unchanged. Since $1/(v+v^2) = 1/v - 1/(1+v)$ has antiderivative $\log(v/(1+v))$, we get $\int_1^\infty dv/(v+v^2) = \log 2$ but $\int_0^1 dv/(v+v^2) = \infty$. So (H4b) holds and (H4a) fails.

When T_* and S_* are both finite, the same classical mechanism applied to $\tilde{f}(y) = f(-y)$ in reversed time gives a two-parameter family: for $(S, T) \in [0, \infty]^2$ there is a solution whose zero set is $[-S, T]$, defined on $(-S - S_*, T + T_*)$, with the convention that an infinite parameter removes the corresponding bound. Each of these is maximal in the

standard sense, and among them $x \equiv 0$ is the only one defined on all of $\mathbb{R}$. We do not claim this family is exhaustive; nothing below needs that, since the point turns on the solutions the family does contain.

The point relevant to Section 5 falls out. For every S , the restriction to $[0, \infty)$ of the solution indexed by (S, T) is the same x_T — the zero set meets $[0, \infty)$ in $[0, T]$ whatever S is. So each forward-maximal solution extends backward in continuum-many ways, and none of them is maximal.

B A mechanical realisation

It is natural to ask whether a Newtonian particle can exhibit the mechanism. It can, though not in the way one first tries.

B.1 Not by steepening a dome

Norton’s construction [14] places a particle on a frictionless surface of revolution, with r the arc length along a meridian from the apex and $h(r)$ the depth below it, giving $\ddot{r} = g h'(r)$. The construction has been discussed at length by Malament [12] and Fletcher [10]; see also Bhat and Bernstein [3]. The obvious candidate here, $h(r) = \frac{1}{g} \left(\frac{2}{3} r^{3/2} + \frac{1}{3} r^3 \right)$, does give $\ddot{r} = \sqrt{r} + r^2$, but it cannot be realised.

What blocks it is a constraint already present in Norton’s own presentation. Writing the tangential force as $g \sin \theta$ with $\sin \theta = dh/dr$ [14] makes $|h'| \leq 1$ mandatory, since a sine is at most 1; the same bound is what an arc-length parametrisation forces, and it is why a dome of this kind extends only to a finite radius. *Neither the bound nor Proposition 12 below is claimed as new.* The proposition is stated only to close one route from the example of Section 2 to a Newtonian realisation.

Proposition 12. *In a Norton-type model — frictionless, uniform gravity, arc-length meridional coordinate, purely meridional motion with zero angular momentum, and the particle remaining on the surface — no surface of revolution admits a solution reaching $r = +\infty$ in finite time.*

Proof. Parametrising the meridian by arc length forces $\rho'^2 + z'^2 = 1$ and hence $|h'| \leq 1$, so $|\ddot{r}| \leq g$. Integrating this bound twice gives $r(t) \leq r(0) + \dot{r}(0)t + \frac{1}{2}gt^2$, which is finite for every finite t . $\square$

With $g = 9.8$ the candidate above already violates the constraint at $r \approx 2.85$. The bound genuinely uses the whole model: with angular momentum the equation gains a centrifugal term $(p_\theta^2/m^2) \rho'(r)/\rho(r)^3$, unbounded as $\rho \rightarrow 0$. So this closes one route and no more.

B.2 In a potential

Let $V(x) = -\left(\frac{2}{3}x^{3/2} + \frac{1}{3}x^3\right)$ for $x \geq 0$, so that $-V'(x) = \sqrt{x} + x^2$. The Newtonian problem is

$$\ddot{x} = \sqrt{x} + x^2, \quad x(0) = \dot{x}(0) = 0. \quad (4)$$

The force is defined only for $x \geq 0$, so by a *solution* of (4) we mean a twice differentiable $x: I \rightarrow [0, \infty)$ on an interval $I \subseteq [0, \infty)$ with $\min I = 0$, satisfying the equation throughout and both initial conditions. Put

$$G(x) = \left(\frac{4}{3}x^{3/2} + \frac{2}{3}x^3\right)^{1/2},$$

so that $\frac{1}{2}G^2 = -V$, and $\frac{d}{dx}\left(\frac{1}{2}G^2\right) = GG' = \sqrt{x} + x^2$ for $x > 0$. Both directions of the correspondence between (4) and $\dot{x} = G(x)$ are needed — one to transport the continuum of branches into the mechanical system, the other to show the mechanical system has no further solutions — so we record them together. The reduction is the standard one for a one-dimensional conservative system, and we claim no priority for it; it is written out because the second direction is what licenses the counts below.

Lemma 13. *Let $I \subseteq [0, \infty)$ be an interval with $\min I = 0$. A function $x: I \rightarrow [0, \infty)$ solves (4) if and only if it solves $\dot{x} = G(x)$, $x(0) = 0$.*

Proof. ($\Rightarrow$) Let x solve (4). Since $x \geq 0$ we have $\ddot{x} = \sqrt{x} + x^2 \geq 0$ on I , so $\dot{x}$ is non-decreasing; with $\dot{x}(0) = 0$ this gives $\dot{x} \geq 0$ throughout, hence x is non-decreasing and $x \geq x(0) = 0$. (So the range constraint $x \geq 0$ is maintained by the equation rather than imposed by it: a solution cannot be carried out of $[0, \infty)$ forward in time.) For the energy, extend V to $\mathbb{R}$ by $V_e = V$ on $[0, \infty)$ and $V_e = 0$ on $(-\infty, 0)$. Then V_e is C^1 : away from 0 this is clear, and at 0 both one-sided derivatives vanish while $V'(u) = -(\sqrt{u} + u^2) \rightarrow 0$ as $u \rightarrow 0^+$, so V_e' is continuous there. Since x is C^1 and takes values in $[0, \infty)$, where $V_e = V$, the chain rule applies and $E = \frac{1}{2}\dot{x}^2 + V_e(x)$ is differentiable with

$$\dot{E} = \dot{x}\ddot{x} + V_e'(x)\dot{x} = \dot{x}(\ddot{x} - (\sqrt{x} + x^2)) = 0.$$

Hence $E \equiv E(0) = \frac{1}{2} \cdot 0^2 + V(0) = 0$, so $\dot{x}^2 = -2V(x) = G(x)^2$. As $\dot{x} \geq 0$ and $G \geq 0$, this gives $\dot{x} = G(x)$; and $x(0) = 0$.

($\Leftarrow$) Let x solve $\dot{x} = G(x)$, $x(0) = 0$. Then $\dot{x}(0) = G(0) = 0$, and x is C^1 because $G \circ x$ is continuous. Since $G \geq 0$ on $[0, \infty)$, x is non-decreasing, so $Z = \{t \in I : x(t) = 0\}$ is an interval containing 0; put $T = \sup Z$. If $Z = I$ then $x \equiv 0$ and $\ddot{x} \equiv 0 = \sqrt{0} + 0^2$, so assume otherwise. On the interior of Z we have $\dot{x} \equiv 0$ and hence $\ddot{x} \equiv 0 = \sqrt{0} + 0^2$. On $I \cap (T, \infty)$ we have $x > 0$, where G^2 is smooth and positive, so G is C^1 and $\ddot{x} = G'(x)\dot{x} = G'(x)G(x) = \sqrt{x} + x^2$. Only $t = T$ remains (when $T \in I$). For $T > 0$ the left derivative of $\dot{x}$ at T is 0, since $\dot{x} \equiv 0$ on $[0, T]$; for $T = 0$ no left derivative is in question, 0 being the left endpoint of I . For the right derivative, note that $u^3 \leq u^{3/2}$

for $u \in [0, 1]$, so $G(u)^2 \leq 2u^{3/2}$ and $G(u) \leq \sqrt{2}u^{3/4}$ there. Hence, on any subinterval of $I \cap (T, \infty)$ on which $x \leq 1$,

$$\frac{d}{dt} x^{1/4} = \frac{1}{4} x^{-3/4} \dot{x} = \frac{1}{4} x^{-3/4} G(x) \leq \frac{\sqrt{2}}{4},$$

and integrating from $T + \varepsilon$ to $T + h$ and letting $\varepsilon \downarrow 0$, where $x(T + \varepsilon) \rightarrow x(T) = 0$ by continuity, gives $x(T + h)^{1/4} \leq (\sqrt{2}/4)h$ for all small $h > 0$. Therefore

$$\frac{\dot{x}(T + h) - \dot{x}(T)}{h} = \frac{G(x(T + h))}{h} \leq \frac{\sqrt{2} (x(T + h)^{1/4})^3}{h} \leq \frac{\sqrt{2} (\sqrt{2}/4)^3 h^3}{h} = \frac{h^2}{16} \xrightarrow{h \rightarrow 0^+} 0,$$

so $\ddot{x}(T)$ exists and equals $0 = \sqrt{x(T)} + x(T)^2$. Hence x solves (4). $\square$

The even extension $\hat{G}(x) = G(|x|)$ — an extension of the vector field, not something derived from a potential on $x < 0$, where V is not defined — is continuous with $\hat{G}(0) = 0$ and $\hat{G} > 0$ off the origin; $G(x) \asymp x^{3/4}$ as $x \rightarrow 0^+$ and $\asymp x^{3/2}$ as $x \rightarrow \infty$, so (H1)–(H3) hold for $\hat{G}$. A solution of $\dot{x} = \hat{G}(x)$ with $x(0) = 0$ is non-decreasing by Step 1 of Theorem 7 and so takes values in $[0, \infty)$, where $\hat{G} = G$; so the solutions of $\dot{x} = \hat{G}(x)$, $x(0) = 0$ are exactly those of $\dot{x} = G(x)$, $x(0) = 0$, and by Lemma 13 exactly those of (4). Everything in Sections 3–4 therefore applies to the Newtonian problem (4) itself — it has exactly one solution defined at every $t \geq 0$, and continuum-many forward-maximal ones — with

$$T_*^{\text{mech}} = \int_0^\infty \frac{dx}{G(x)} = \frac{2}{3} \sqrt{\frac{3}{2}} 2^{-1/3} B\left(\frac{1}{6}, \frac{1}{3}\right) \approx 5.4521363617.$$

The closed form is obtained as follows, and the reader can check every step. Factor $G^2 = \frac{4}{3}x^{3/2}\left(1 + \frac{1}{2}x^{3/2}\right)$ and substitute $s = \frac{1}{2}x^{3/2}$, so that $x = (2s)^{2/3}$ and $dx = \frac{2}{3}2^{2/3}s^{-1/3}ds$. Then

$$\frac{dx}{G(x)} = \frac{2}{3} \sqrt{\frac{3}{2}} 2^{-1/3} s^{-5/6} (1 + s)^{-1/2} ds,$$

and since $\int_0^\infty s^{a-1}(1 + s)^{-a-b} ds = B(a, b)$ with $a = \frac{1}{6}$ and $b = \frac{1}{3}$, the integral is $\frac{2}{3} \sqrt{\frac{3}{2}} 2^{-1/3} B\left(\frac{1}{6}, \frac{1}{3}\right)$ as stated.

So the counts of Theorem 7 hold for the Newtonian problem (4) — but for a different member of the class, not for the example of Section 2. The reduced right-hand side G is not $|x|^{2/3} + x^2$, and accordingly $T_*^{\text{mech}} \neq T_*$. Two caveats. The potential is unbounded below; finite-time escape in Newtonian systems, and the responses available to it, are discussed by Earman [8]. And V is C^1 but not C^2 at the origin, the force being continuous and not locally Lipschitz there. Whether such a force law counts as a legitimate Newtonian system is the question the dome literature asks about $\sqrt{r}$ [10, 14]; we take no position.

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